Data Communication and Networks
Cyber Physical Systems

JAIN (Deemed-to-be University)

- *"Education is the most powerful weapon for changing the world"*

- **Nelson Mandela**

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A++ grade CGPA of 3.71/4

68th Rank

AAAA+ Rating

AY- 2021 | 5th In Bengaluru 46th in India

MODULE 1: Introduction to Data Communications – 10 Hours

1.1. Introduction to Data Communication - Components, Data Representation, Data Flow,

1.2. Networks- Distributed Processing, Network Criteria, Physical Structures, Network Models, Categories of Networks, Interconnection of Networks, The Internet – A Brief History, The Internet Today,

1.3 Network Models, Layered Tasks, OSI model, Layers in OSI model, TCP/IP Protocol Suite, Addressing Introduction,

1.4. Wireless Links and Network Characteristics, WiFi: 802.11- Wireless LANs - The 802.11 Architecture.

Wireless Links and Network Characteristics

- Wireless Network: “Wireless” means without wire, media that is made up of electromagnetic waves (EM Waves) or infrared waves.
- Antennas or sensors will be present on all wireless devices. Cellular phones, wireless sensors, TV remotes, satellite disc receivers, and laptops with WLAN cards are all examples of wireless devices.
- For data or voice communication, a wireless network uses radiofrequency waves rather than wires.

Wireless Links and Network Characteristics

1 Wireless Links

Wireless links facilitate communication between devices without direct physical connections. These links operate based on different **frequencies, protocols, and transmission technologies**.

Types of Wireless Links:

1. **Radio Waves** – Used in Wi-Fi, cellular networks, and Bluetooth.
2. **Microwaves** – Used in satellite communication and point-to-point links.
3. **Infrared (IR)** – Used in remote controls and short-range communication.

Wireless Links and Network Characteristics

Wireless Communication Types:

- **Wi-Fi (Wireless Fidelity):** Standardized under **IEEE 802.11**, used in home and office networks.
- **Bluetooth:** Short-range communication for connecting peripherals (e.g., headphones, smart devices).
- **Cellular Networks:** 3G, 4G LTE, 5G used for mobile communication.
- **Satellite Communication:** Used in GPS, broadband services, and remote area connectivity.
- **NFC (Near Field Communication):** Short-range wireless for contactless payments and authentication.

Wireless Links and Network Characteristics

Key Characteristics:

1. **Mobility** – Wireless devices can move freely while remaining connected.
2. **Flexibility** – Easy to deploy without complex cabling.
3. **Scalability** – Can expand coverage by adding access points or range extenders.
4. **Interference** – Signal disruption due to obstacles, other devices, or environmental factors.
5. **Latency** – Wireless networks often have **higher delays** compared to wired connections.
6. **Security** – Wireless links are more vulnerable to **eavesdropping, hacking, and interference** than wired networks.
7. **Bandwidth** – Limited compared to fiber optics, but improved with **5G and Wi-Fi 6**.
8. **Reliability** – Subject to fluctuations due to weather conditions, interference, or device congestion.

Wireless Links and Network Characteristics

3 Wireless Network Categories

Wireless networks can be categorized based on range, coverage, and usage.

Network Type	Range	Example
WPAN (Wireless Personal Area Network)	A few meters	Bluetooth, ZigBee
WLAN (Wireless Local Area Network)	100-200m	Wi-Fi (IEEE 802.11)
WMAN (Wireless Metropolitan Area Network)	Several km	WiMAX (IEEE 802.16)
WWAN (Wireless Wide Area Network)	10+ km	Cellular Networks (3G, 4G, 5G)
Satellite Networks	Global	GPS, Broadband

Wi-Fi: IEEE 802.11 / WLAN

- **1. What is Wi-Fi?**
- Wi-Fi is a wireless networking technology that allows devices like laptops, smartphones, and tablets to connect to the internet without cables.
- It is based on the IEEE 802.11 family of standards.
  - Uses radio waves to transmit data
  - Works within a limited area (home, office, campus, café)
  - Provides internet access through a wireless router or access point.

Types of Wi-Fi Connection

1. Wi-Fi router
2. 4G LTE home internet
3. Mobile hotspot
4. 5G home internet

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Wi-Fi: IEEE 802.11 and Wireless LAN

1 IEEE 802.11 Standards

The **IEEE 802.11 family** defines specifications for WLANs. The key **802.11 standards** include:

Standard	Frequency Band	Max Speed	Range	Key Features
802.11a	5 GHz	54 Mbps	30m	Faster but shorter range than 802.11b
802.11b	2.4 GHz	11 Mbps	100m	First widely adopted Wi-Fi standard
802.11g	2.4 GHz	54 Mbps	100m	Backward compatible with 802.11b
802.11n	2.4 & 5 GHz	600 Mbps	200m	Uses **MIMO** for higher speeds
802.11ac	5 GHz	1.3 Gbps	200m	Supports **MU-MIMO**, higher bandwidth
802.11ax (Wi-Fi 6)	2.4 & 5 GHz	9.6 Gbps	250m	Increased efficiency, lower latency
802.11be (Wi-Fi 7)	2.4, 5 & 6 GHz	40+ Gbps	250m+	Ultra-high-speed, improved gaming & streaming

1.5. WiFi: 802.11- Wireless LANs - The 802.11 Architecture.

Wi-Fi Evolution Timeline

1999 — 802.11b

Speed: Up to 11 Mbps
Frequency: 2.4 GHz
Range: Moderate, suitable for indoor and outdoor use
Key Features: Wider range, lower cost, and higher compatibility than 802.11a

1999 — 802.11a

Speed: Up to 54 Mbps
Frequency: 5 GHz
Range: Short, primarily indoor
Key Features: High data rate, less prone to interference

2003 — 802.11g

Speed: Up to 54 Mbps
Frequency: 2.4 GHz
Range: Moderate, suitable for indoor and outdoor use
Key Features: Improved speed compared to 802.11b, backward compatibility with 802.11b devices

2009 — 802.11n (Wi-Fi 4)

Speed: Up to 600 Mbps
Frequency: 2.4 GHz and 5 GHz
Range: Extended, suitable for indoor and outdoor use
Key Features: MIMO technology, improved range and speed, increased capacity, and better power management

2013 — 802.11ac (WI-FI 5)

Speed: Up to 3.46 Gbps
Frequency: 5 GHz
Range: Extended, suitable for indoor and outdoor use
Key Features: MU-MIMO technology, faster data rates, improved network efficiency, and support for more connected devices

2019 — 802.11ax (Wi-Fi 6)

Speed: Up to 9.6 Gbps
Frequency: 2.4 GHz and 5 GHz
Range: Extended, suitable for indoor and outdoor use
Key Features: OFDMA, MU-MIMO, increased capacity, faster data rates, improved efficiency, and better power management with Target Wake Time (TWT)

(Wi-Fi 7)
Coming next

Wi-Fi: IEEE 802.11 and Wireless LAN Architecture

2 802.11 WLAN Architecture

A **Wireless LAN (WLAN)** consists of various components that ensure seamless communication between devices.

Components of 802.11 WLAN

1. **Station (STA):** Any device (laptop, phone, IoT device) with a wireless adapter.
2. **Access Point (AP):** Connects wireless devices to a wired network.
3. **Basic Service Set (BSS):** A group of stations connected to an AP.
4. **Extended Service Set (ESS):** Multiple APs connected to form a larger network.
5. **Distribution System (DS):** Connects multiple APs and extends coverage.

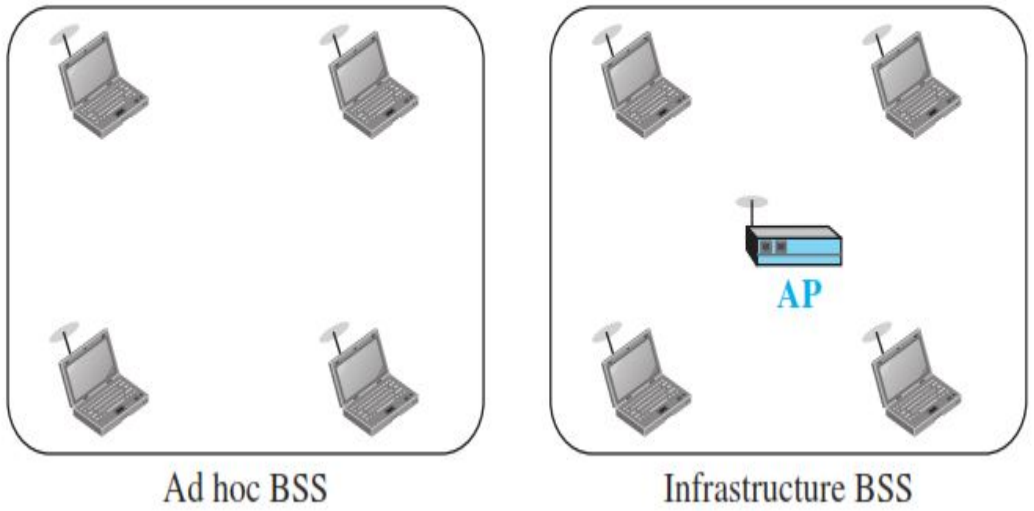
Wi-Fi: IEEE 802.11 / Wireless LAN Architecture

The standard defines two kinds of services: the basic service set (BSS) and the extended service set (ESS).

1. Basic Service Set

- IEEE 802.11 defines the basic service set (BSS) as the building blocks of a wireless LAN.
- A basic service set is made of stationary or mobile wireless stations and an optional central base station, known as the access point (AP).

Figure 15.4 *Basic service sets (BSSs)*



- The BSS without an AP is a stand-alone network and cannot send data to other BSSs. It is called an ad hoc architecture. In this architecture, stations can form a network without the need of an AP; they can locate one another and agree to be part of a BSS.
- A BSS with an AP is sometimes referred to as an infrastructure BSS.

Wi-Fi: IEEE 802.11 and Wireless LAN Architecture

2. Extended service set (ESS)

- An extended service set (ESS) is made up of two or more BSSs with APs. In this case, the BSSs are connected through a distribution system, which is a wired or a wireless network.
- The distribution system connects the APs in the BSSs. IEEE 802.11 does not restrict the distribution system; it can be any IEEE LAN such as an Ethernet.

Station Types

IEEE 802.11 defines three types of stations based on their mobility in a wireless LAN:

1. **no-transition -** A station with no-transition mobility is either stationary (not moving) or moving only inside a BSS
2. **BSS-transition -** A station with BSS-transition mobility can move from one BSS to another, but the movement is confined inside one ESS
3. **ESS-transition mobility -** A station with ESS-transition mobility can move from one ESS to another. However, IEEE 802.11 does not guarantee that communication is continuous during the move.

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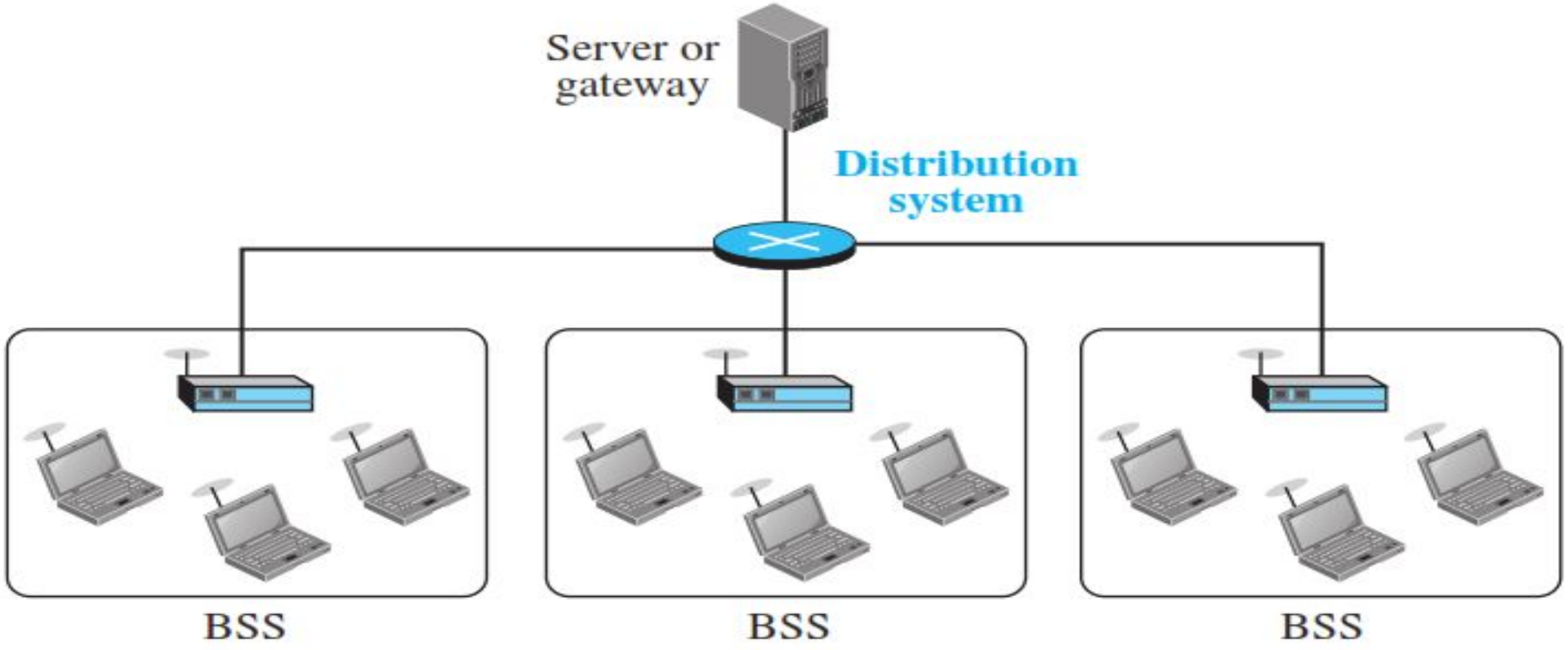
Wi-Fi: IEEE 802.11 and Wireless LAN Architecture

Figure 15.5 *Extended service set (ESS)*



Wireless Links and Network Characteristics

4 Advantages and Disadvantages of Wireless Networks

✅ Advantages:

- ✔ **Easy Installation** – No need for physical cables.
- ✔ **Supports Mobility** – Devices can connect on the move.
- ✔ **Scalability** – Can extend coverage with additional access points.
- ✔ **Cost-Effective** – No infrastructure costs for wiring.

❌ Disadvantages:

- ❌ **Security Risks** – Susceptible to hacking and unauthorized access.
- ❌ **Signal Interference** – Affected by walls, weather, and other devices.
- ❌ **Limited Bandwidth** – Slower than wired networks in high-traffic scenarios.
- ❌ **Power Consumption** – Wireless devices often consume more power than wired connections.

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thank you